

VerityNgn OSS 3.0.0 — Release Paper

VerityNgn OSS 3.0.0 — Public Engine Release Paper

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License: Apache-2.0 (engine); commercial overlays under separate terms

Abstract

We release **VerityNgn OSS 3.0.0**, a standalone open-source multimodal engine for claim extraction and verification from video. This release restores **Deep Research** (grounded forensic summarization) to the public Apache tree, adds optional **authenticity** and **spectral voice** cues, and documents a clear open-core boundary: predictions and RiskFactor products remain closed overlays that **depend on OSS**, not the reverse. We do not claim earnings prediction, allocation factors, or lie detection.

Keywords: video verification, counter-intelligence, deep research, open-core, multimodal LLM

1. Motivation

Prior open-core practice left Deep Research and several authenticity cues on the commercial fork, while OSS drifted. Public distribution requires an impressive, drop-in engine (``pip install verityngn``) that communities can run without SaaS. Overlays for hosted multi-tenant delivery, predictions, and RiskFactor must build **on** OSS.

2. System overview (v3)

```

Video URL or file

- multimodal analysis + claim extraction
- counter-intelligence (YouTube / press-release / Sherlock CI)
- probabilistic verification (TRUE / FALSE / UNCERTAIN)
- standard report (MD / HTML / JSON / PDF)
- optional Deep Research (Gemini grounded pass + combined report)
- optional authenticity / spectral / Face Landmarker cues

```

2.1 Deep Research (now in OSS)

``verityngn analyze <url> --deep`` runs a counter-intel sanitize of ``{video_id}_report.json``,

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- grounding audit JSON
- combined TL;DR + DR + standard report when available

2.2 Authenticity & vision (optional extras)

- Authenticity gate with stub adapters (C2PA / Corsound / visual) — fail-closed when checks fail
- Spectral AI-voice indicators via optional `librosa`
- MediaPipe Face Landmarker adapter for congruence / delivery-risk cues only (not employment use)

2.3 Explicitly out of this release

Predictions signal engine, CAR/ICIR, Karp panels, RiskFactor S00–S15, hosted trial credits, full Gemini multisense fusion (deferred).

3. Evaluation narrative (illustrative)

Reported evaluation on prior OSS corpora emphasized counter-intelligence lift and calibrated three-state verdicts. Illustrative charts (not predictions metrics):

Live operator parity vs commercial core: see `PARITY\_TEST\_v3.md`.

4. Install & CLI

```
```bash
pip install verityngn
or: pip install -e ".[deep,vision,audio]"

verityngn analyze https://www.youtube.com/watch?v=VIDEO_ID
verityngn analyze <url> --deep
verityngn analyze --file deposition.mp4 --title "Matter clip"
```
```

Credentials: Vertex ADC or `VERITY\_GEMINI\_KEY` / `GEMINI\_API\_KEY` (see `.env.example`).

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5. Open-core dependency model

Future commercial, predictions, and RiskFactor repositories pin **OSS ≥3.0.0** and keep overlay-only packages. Boundary CI fails if ``services/predictions`` or ``services/quant`` appear under the public package.

See ``docs/OPEN_CORE.md`` and ``docs/DEPENDENCY_MODEL.md``.

6. What we will not claim

- No earnings prediction or allocation alpha
- No “lie detector” branding
- No RiskFactor registry scores in this paper

7. Related documents

- Full methodology paper: ``papers/verityngn_research_paper.md`` (v2 lineage; updated by this release note)
- Counter-intelligence: ``papers/counter_intelligence_methodology.md``
- Probability model: ``papers/probability_model_foundations.md``
- Launch pack: ``docs/launch/``

Changelog pointer

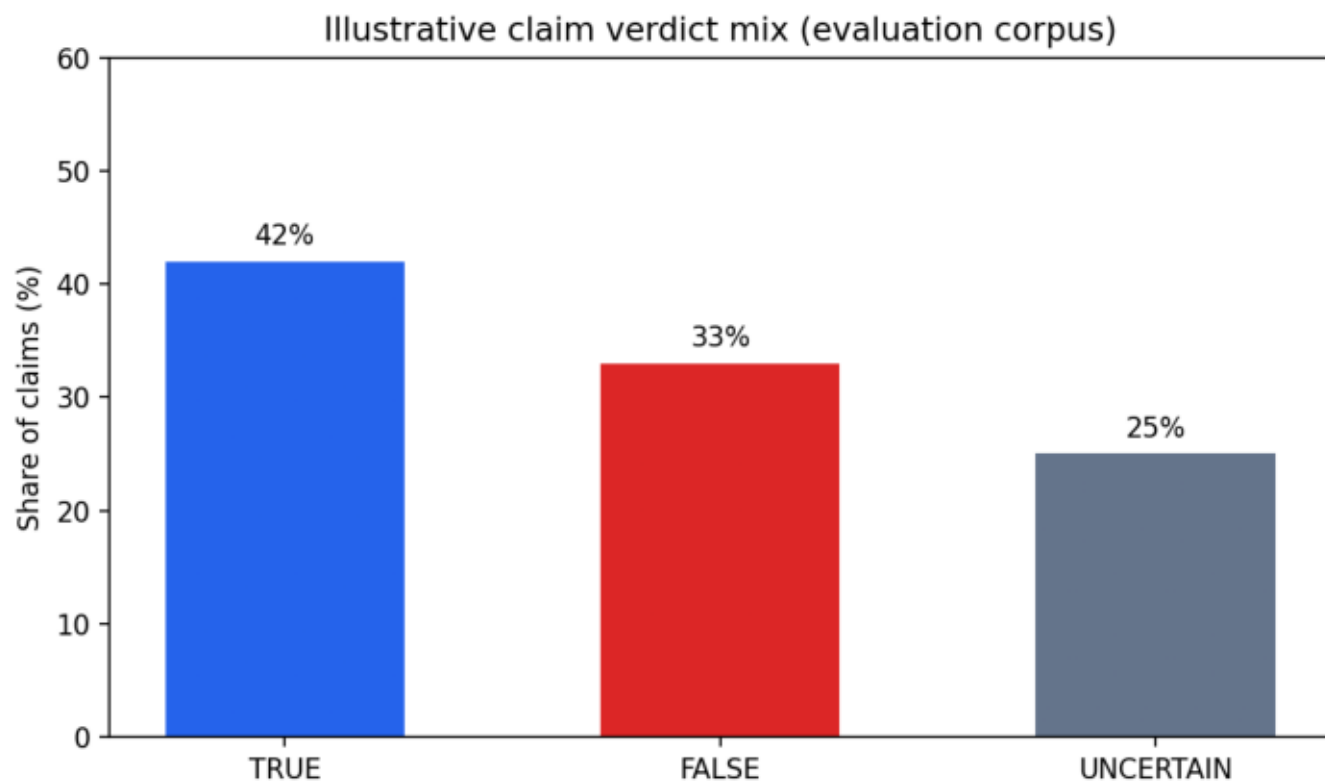
See repository ``CHANGELOG.md`` → ``[3.0.0] - 2026-08-29``.

open_core_architecture.png

VerityNgn open-core (v3.0.0)



claim_verdict_mix.png



accuracy_stages.png

